

Configuration of Network address translation in Cisco packet tracer



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Network Address Translation

- Network address translation (NAT) is a technique that allows multiple devices to share a single IP address.
- NAT is often used by internet service providers (ISPs) and organizations.
- **How NAT works?**
 - A router intercepts data sent from a private network to a public network
 - The router replaces the private IP address with its own public IP address
 - The router sends the data to the destination device
 - When data is sent back to the router, the router replaces the public IP address with the original private IP address
- **Benefits**
 - Conserves public IP addresses
 - Provides security by hiding devices' actual IP addresses



Types of NAT

- Static NAT

- IP addresses are statically mapped to each other through manual configuration.
- Global IP addresses are translated to Local IP addresses based on the statically mapping of these IP addresses.
- <https://www.geeksforgeeks.org/static-nat-configuration-in-cisco/>

- Dynamic NAT

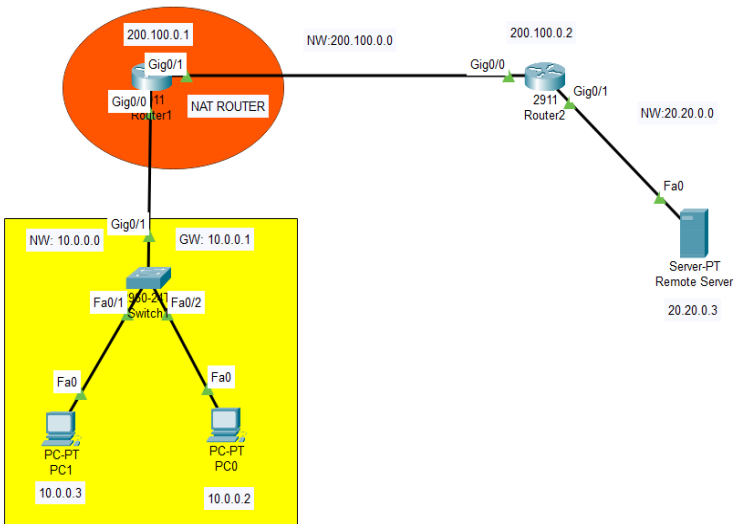
- IP addresses are dynamically mapped to each other on a one-to-one basis as per the needs.
- This type of translation is very useful when there are multiple users in a private network that access the Internet.
- <https://www.geeksforgeeks.org/dynamic-nat-configuration-in-cisco/?ref=lbp>

- The three reserved ranges are:

- 10.0.0.0 – 10.255.255.255/8 (16,777,216 hosts)
- 172.16.0.0 – 172.31.255.255/12 (1,048,576 hosts)
- 192.168.0.0 – 192.168.255.255/16 (65,536 hosts)



NAT Implementation (Static)





Devices and Connections

- Select two routers (Router-2911) and place them on the workspace
- Select one switch (Switch-2960) and place it on the workspace
- Select two PCs and place them on the workspace
- Select one server (Server-PT) and place it on the workspace
- Connections for left-side network
 - Use a Copper straight-through and connect
 - FastEthernet of PC0 $\Rightarrow$ FastEthernet0/2 of the switch
 - FastEthernet of PC1 $\Rightarrow$ FastEthernet0/1 of the switch
 - GigabitEthernet0/1 of switch to GigabitEthernet0/0 of the left-side router
- Connections for right-side network
 - Use a Copper straight-through and connect
 - GigabitEthernet0/1 of Router1 $\Rightarrow$ GigabitEthernet0/0 of the Router2
 - FastEthernet of Server $\Rightarrow$ GigabitEthernet0/1 of the Router2



Assign IP Address

Click on PC → Desktop Tab → IP Configuration

- PC0: IPv4: 10.0.0.2; SubMask:255.0.0.0; Gateway: 10.0.0.1
- PC1: IPv4: 10.0.0.3; Sub Mask:255.0.0.0; Gateway: 10.0.0.1
- Server: IPv4: 20.20.0.3; Sub Mask:255.0.0.0; Gateway: 20.20.0.1



Configuring Gateway of router (Left)

Router2

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0**
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000C.8503.6901

IP Configuration

IPv4 Address 10.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ip address 10.0.0.1 255.0.0.0
Router(config-if)#
```



Configuring Interface of the router (Left)

Router2

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1**
- GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 000C.8503.6902

IP Configuration

IPv4 Address 200.100.0.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ip address 10.0.0.1 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
ip address 200.100.0.1 255.255.255.0
Router(config-if)#
```




Configuring Gateway of the router (Right)

Router3

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1**
- GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0006.2A16.5A02

IP Configuration

IPv4 Address 20.20.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 20.20.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```



Configuring Interface of the router (Right)

Router3

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0**
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps

Duplex ☐ Half Duplex ☒ Full Duplex

MAC Address 0006.2A16.5A01

IP Configuration

IPv4 Address 200.100.0.2

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ip address 200.100.0.2 255.255.255.0
Router(config-if)#
```



RIP Configuration of the router (left)

We can use Static, RIP or OSPF routing. RIP is shown here.

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#network 200.100.0.0
Router(config-router)#exit
Router(config)#exit
n.....
```



RIP Configuration of the router (Right)

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]? t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 200.100.0.0
Router(config-router)#network 20.20.0.0
Router(config-router)#exit
Router(config)#exit
```



Check Connections before NAT

- Ping the PCs to check connection. Every PC is connected to every other PC
- Inspect the Routing table
- "sh ip route rip" in the CLI of the router



NAT Configuration

• NAT Configuration

- Router(config)#int g0/0
- Router(config-if)#ip nat inside
- Router(config-if)#exit
- Router(config)#int g0/1
- Router(config-if)#ip nat outside
- Router(config-if)#exit

• Mapping

- Router(config)#ip nat inside source static 10.0.0.3 200.100.0.1
- After ping from both PCs check NAT translations
- Router#sh ip nat translations



NAT Translations

For detailed explanation of outside local/global:

https:

[//www.geeksforgeeks.org/static-nat-configuration-in-cisco/](https://www.geeksforgeeks.org/static-nat-configuration-in-cisco/)

```
Router#sh ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	200.100.0.1:16	10.0.0.2:16	20.20.0.3:16	20.20.0.3:16
icmp	200.100.0.1:17	10.0.0.2:17	20.20.0.3:17	20.20.0.3:17
icmp	200.100.0.1:18	10.0.0.2:18	20.20.0.3:18	20.20.0.3:18
icmp	200.100.0.1:19	10.0.0.2:19	20.20.0.3:19	20.20.0.3:19
icmp	200.100.0.1:23	10.0.0.2:23	20.20.0.3:23	20.20.0.3:23
icmp	200.100.0.1:24	10.0.0.2:24	20.20.0.3:24	20.20.0.3:24
icmp	200.100.0.1:25	10.0.0.2:25	20.20.0.3:25	20.20.0.3:25
icmp	200.100.0.1:26	10.0.0.2:26	20.20.0.3:26	20.20.0.3:26
icmp	200.100.0.1:27	10.0.0.2:27	20.20.0.3:27	20.20.0.3:27
icmp	200.100.0.1:28	10.0.0.2:28	20.20.0.3:28	20.20.0.3:28
---	200.100.0.1	10.0.0.3	---	---



Observing NAT translations

To observe NAT translations: Enter "debug ip nat" in the Router terminal and ping the server from one of the PCs

```
Router#debug ip nat
IP NAT debugging is on
Router#
NAT: s=10.0.0.2->200.100.0.1, d=20.20.0.3 [34]
NAT*: s=20.20.0.3, d=200.100.0.1->10.0.0.2 [25]
NAT: s=10.0.0.2->200.100.0.1, d=20.20.0.3 [35]
NAT*: s=20.20.0.3, d=200.100.0.1->10.0.0.2 [26]
NAT: s=10.0.0.2->200.100.0.1, d=20.20.0.3 [36]
NAT*: s=20.20.0.3, d=200.100.0.1->10.0.0.2 [27]
NAT: s=10.0.0.2->200.100.0.1, d=20.20.0.3 [37]
NAT*: s=20.20.0.3, d=200.100.0.1->10.0.0.2 [28]
```

```
C:\>ping 20.20.0.3
```

```
Pinging 20.20.0.3 with 32 bytes of data:
```

```
Reply from 20.20.0.3: bytes=32 time=1ms TTL=126
Reply from 20.20.0.3: bytes=32 time<1ms TTL=126
Reply from 20.20.0.3: bytes=32 time<1ms TTL=126
Reply from 20.20.0.3: bytes=32 time<1ms TTL=126
```

```
Ping statistics for 20.20.0.3:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```




Conclusion

Configuring static NAT to map a private internal IP to a specific public external IP.

Configuring static NAT is essential for scenarios where a device inside the private network needs to be accessible from the outside world using a fixed public IP address.